

# WARIT YUVANIYAMA (NOTE)

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Fourth Year Computer Engineering Student  
Sirindhorn International Institute of Technology (SIIT), Thammasat University



## SUMMARY

High-achieving fourth-year Computer Engineering student (3.92 CGPA) and President of SIIT Robotic Club (OrcaBOT). Proficient in **embedded systems design** using **C++**, **Python**, and **ROS2**, with peer-reviewed research publications. Proven leader in managing technical procurement (budgeting **100,000+ THB**) and an award-winning communicator with a track record in technical pitching for innovative solutions in EdTech and Sustainability.

## EDUCATION

**Bachelor of Computer Engineering**, SIIT, Thammasat University Expected 2027  
CGPA: 3.92 (Expected First-Class Honors); Recipient of Academic Outstanding Student Scholarship.

- *Relevant Coursework*: Microcontrollers & Applications, Database Systems, Operating Systems, Computer Architecture, Artificial Intelligence & Applications, Object-Oriented Programming, Algorithm Design.

## SKILLS

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|-------------------------|--------------------------------------------------------------------------------------|
| <b>Programming</b>      | Python, C++, Java, ROS2, micro-ROS, Basic Assembly, LaTeX                            |
| <b>Systems &amp; DB</b> | Relational Database Design, SQL (MySQL), Ubuntu Linux, Express.js                    |
| <b>Hardware</b>         | ESP32, Raspberry Pi, Arduino, KiCad, Sensors/Actuators, MATLAB                       |
| <b>Technical</b>        | Project Management, Research & Technical Writing, Technical Pitching                 |
| <b>Languages</b>        | Thai (Native), English (Fluent)   SAT: 1420 (M:790, EBRW:630), TOEFL (ITP): 553 (B2) |

## EXPERIENCE

**Research Intern, Measurement & Intelligence Systems Lab, Kansai University, Japan** Jun–Jul 2026

- Developed a **mmWave radar + thermal imaging sensor fusion** pipeline for autonomous robot navigation in smoke-filled, low-visibility environments where RGB cameras and LiDAR degrade (TI IWR1443, FLIR Boson 320 on TurtleBot3).
- Built the full **ROS2 Humble** integration: custom sensor drivers, hardware-accelerated thermal frame streaming, TF-tree extrinsics, calibration, and **ApproximateTimeSynchronizer**-based multi-sensor fusion overlaying depth-colored radar point clouds in real time.

**President & Embedded Systems Lead, SIIT Robotic Club (OrcaBOT)** Oct 2023 – Present

- **Leadership**: Leading a 30-member club, managing a **100,000+ THB** procurement budget, and competing in **RoboCup Japan Open 2025** with an automatic soccer robot.
- **Firmware Architecture**: Developed custom C++ motor-control libraries and holonomic drive algorithms; redesigned serial and wireless nRF communication systems to significantly increase responsiveness and stability.
- **Technical Communication**: Primary author of the Software Architecture section (System Overview, Low-/High-Level Design) of the RoboCup 2024 Team Description Paper. [\[Link\]](#)

**Intern, NECTEC, NSTDA** May 2024 – Dec 2025

- Implemented Fast Segment Anything Model (FastSAM) for a water surface level monitoring system; built a scaled wave-tank coastal erosion test model.
- Presented research findings at international venues, including IUKM 2025 (Vietnam).

## RESEARCH PUBLICATIONS

- Yuvaniyama, W. (*First Author*), *et al.* (2026). Thai Poetry Rhyme Verification Using a Hybrid Rule-Based Approach for Education and GenAI. *iSAI-NLP 2026* (In review). *Rhyme verifier merged into PyThaiNLP; reward signal for GenAI Klon-Paed generation.*

- Chetsawang, J., **Yuvanityama, W.**, Boonkwan, P. (2027). Direction-Aware Reward Shaping for Fuel-Efficient Orbital Transfer using Reinforcement Learning. *IEICE Trans. Communications, E110-B(6)*. *~10% fuel reduction via physics-informed PPO shaping*.
- Chetsawang, J., **Yuvanityama, W.**, et al. (2027). Online Rapid Innate Learning via Recursive Least-Squares for Legged Locomotion. *IEEE/SICE SII 2027*, Kobe, Japan. *Heading error 26.3° → 5.5° on Unitree Go2*.
- Chetsawang, J., **Yuvanityama, W. (Presenter)**, et al. (2025). Water Surface Level Estimation Based on the Fast Segment Anything Model for Coastal Monitoring Systems. *IUKM 2025, LNCS vol 15585, Springer*. [DOI]
- Thongsupan, T., Sureechainirun, N., **Yuvanityama, W.**, et al. (2024). OrcaBOT Team Description. *RoboCup 2024 Symposium TDPs — Small Size League*. [Link]

## PROJECTS

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- **FlashLight: Companion Study Pacer Application** *Co-Founder, Operations Lead & Pitcher*  
Co-founded FlashLight ([flashlight.in.th](https://flashlight.in.th)), a study-planning platform pacing Thai students' TCAS exam prep by breaking 368+ A-Level topics into personalized daily missions. Delivered the pitch in every elimination round; reached **Top 5** (of 80+ teams) with an iOS MVP ranked **#7 in Education** on the Thai App Store.
- **CareAir: Air Quality Monitoring & Carbon Capture System (Hackathon Winner)**  
Developed "CareAir", a proof-of-concept carbon capture machine designed to improve air quality in classrooms/workplaces. Awarded **1st Place** (Physical Wellness Track) at Thammasat Hackathon: Future Wellness 2024.
- **CrystalEyes: Automated Crystal Imaging System**  
Developed a low-cost (10k–15k THB) CNC-based digital microscope system to automate crystallization screening, replicating core functionality of commercial systems costing 1M+ THB. Iteratively designed and built from high school through university; demonstrated with lysozyme growth and MOF studies.  
Links: ([Video](#)) — ([Report/Poster](#))
- **Autolocate: Condo Parking Management System**  
Designed a relational database backend using **SQL and Express.js** to manage high-traffic condo parking operations, implementing a three-tier connection pool system to enforce role-based access control (RBAC) and ensuring secure authentication via JWT and HttpOnly cookies.  
Link: ([GitHub Repository](#))

## WORKSHOPS & INTERNATIONAL PROGRAMS

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- **Staff, PBL Workshop: Automatic Robot in Industrial Conveyor Belt Challenge** August 2026  
Planned and organized the event, built the TA demonstration robot, and supervised each participant team. Theme: multi-sensor navigation using **IMU, wheel encoders, and ultrasonic sensors with PID control** for robot heading identification and navigation.
- **Staff, International Project-Based Learning (iPBL) Workshop: Rescue Robot** August 2025  
Managed technical logistics and material procurement for teams from SIIT and OIT Japan. Provided hands-on mentorship in robot design and debugging, specializing in ROS2, micro-ROS, and Linux integration on Raspberry Pi and ESP32 platforms, while facilitating cross-cultural collaboration between Thai and Japanese students.
- **Participant, 11th Problem-Based Learning (PBL) Workshop, Taipei Tech, Taiwan** August 2024  
Collaborated within an international team (12 universities represented) to design and build a smart autonomous vehicle using Arduino, sensors/actuators, and 3D modeling for color recognition, positioning, and object grasping tasks over 8 days.

## AWARDS AND RECOGNITION

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- **3rd Runner-Up (Top 5)**, Chulalongkorn University Startup Thailand League (CUSTL) — FlashLight Team  
Selected as an **official representative of Chulalongkorn University** for the Startup Thailand League 2026 **Regional Pitching Round (Central Thailand)**.
- **Qualified & Exhibited**, SITE2026 Demo Day (NIA, Paragon Hall) — Startup Thailand League 2026  
Pitched FlashLight to VCs and investors from Thailand, South Korea, and beyond.
- **1st Place Winner**, Thammasat Hackathon: Future Wellness 2024 (CareAir Team)
- **Finalist**, ASEAN-China-India Youth Leadership Summit 2024 Sustainability Startathon (Singapore)
- **3rd Place (of 200 teams)**, 5th Kibo Robot Programming Challenge Thailand — developed autonomous navigation and **ArUco marker detection in Java** for ISS Astrobee robots.